Can we predict which YouTube Shorts will go viral?

Erdős Institute
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Introduction

Aim: Identify early indicators of YouTube Shorts virality and assess whether we can predict it within the first **six hours**.

Viral Video: 72-hour views exceed the top-10% threshold

Unit of analysis: one short

Stakeholders:

- Creators, creator agencies, brands, advertisers
- YouTube’s trend and curation teams

KPIs:

- Average precision (PR AUC)
- ROC AUC

XXXX

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- Creators, creator agencies, brands, advertisers
- YouTube’s trend and curation teams could use early predictions to prioritize content.

KPIs:

- Average precision (AP), with random-ranking baseline of approximately 0.10 when 10% of videos are viral.
- ROC AUC, plus precision and recall when flagging the top 10% of model scores.

Data

- **Source**: YouTube API, polled every hour by a GitHub Actions Collector
- **Scale**:
  - Collected data from Oct 12 to Nov 1
  - 30,709 videos
  - 15,687 channels
  - 2.17M snapshots in 440 hourly files
- **Collection quality**:
  - Median gap between snapshots: 1.0 hr
  - 94% of videos have data at 72 hr

File	One row per	Contents
channels.csv	channel	title, creation date, subscribers, total views/videos (as of first sighting)
videos.csv	video	publish time, channel, title, description, duration, tags
channel_countries.csv	channel	self-reported channel country
observations.csv	video x snapshot	view count, like count, comment count

EXPLORATORY DATA ANALYSIS

EDA - Correlation Matrix

Strong Early Growth Correlation

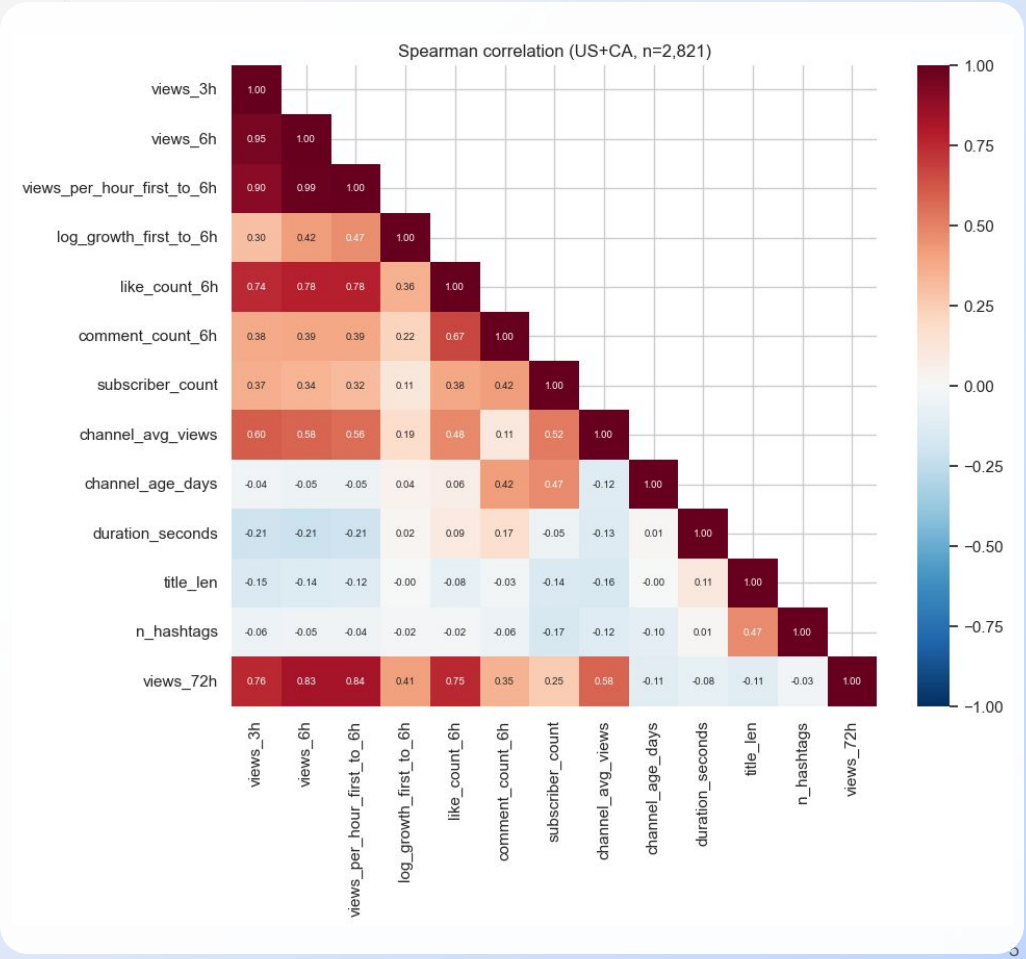
The early views are highly correlated with growth; the more views a video receives within the first few hours, the more views the videos ends up having at the end of 72 hours.

Positive Signals

Comments, subscriber count, and the like count also have positive correlations with the ending view count at 72 hours

Negative Correlation

The duration and title length have a weak negative correlation with the ending view count after 72 hours.



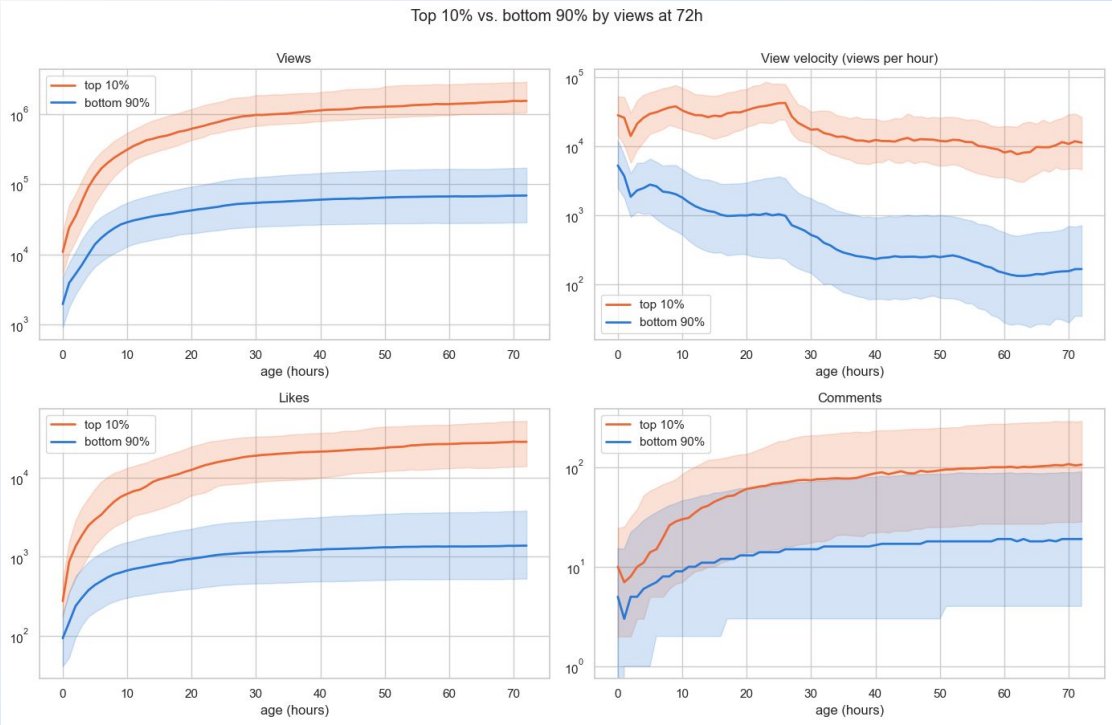
EDA - Trajectories

10-Hour Horizon

After the first 10 hours, growth slows but viral videos keep gaining views.

24-Hour Velocity Drop

After the first 24 hours, the views per hour plummets significantly for viral videos



FEATURE ENGINEERING

What the model knew by six hours

Predictors were grouped by source and restricted to information available by the forecasting cutoff.

01 Early performance

log(1 + views at 6 h)
log-view growth from 3h to 6h

02 Engagement

Raw likes and comments at 6h
Indicator for hidden or missing likes

03 Creator context

Subscribers, channel age and historical channel scale
Masked if unavailable by 6h

04 Content + timing

Duration, hashtags, title length and emojis
Publication hour encoded cyclically

05 Alternative encoding

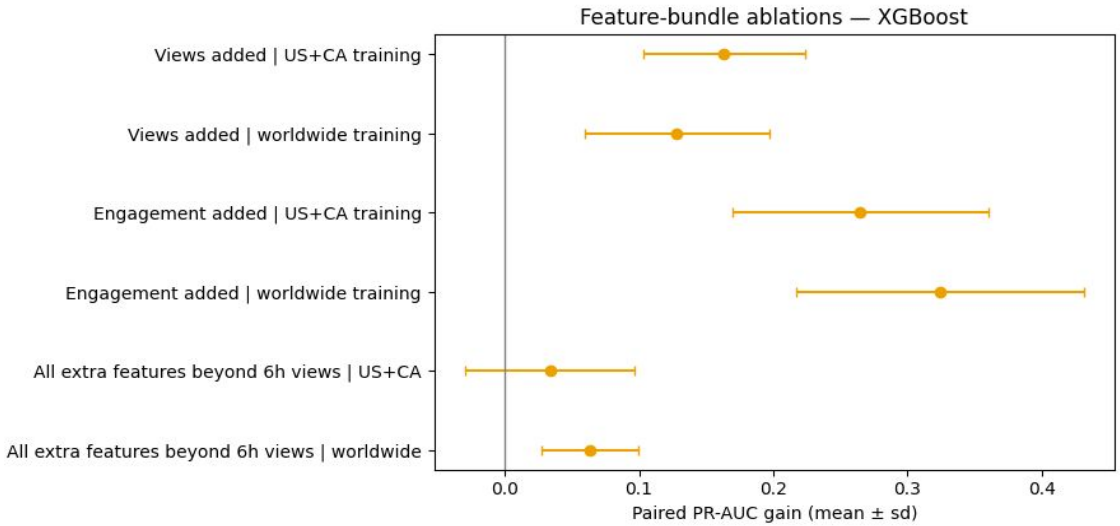
Raw versus log1p likes/comments compared on identical CV folds

Availability rules snapshot at or before 6 h · missing likes not set to zero · no view-normalized engagement rates

FEATURE ENGINEERING

Paired ablations clarify what each feature bundle adds

Positive values indicate higher PR-AUC when the named bundle is added on the same CV folds.



Development evidence

+0.324

engagement added to metadata (worldwide training)

+0.128

early views added to the no-view bundle

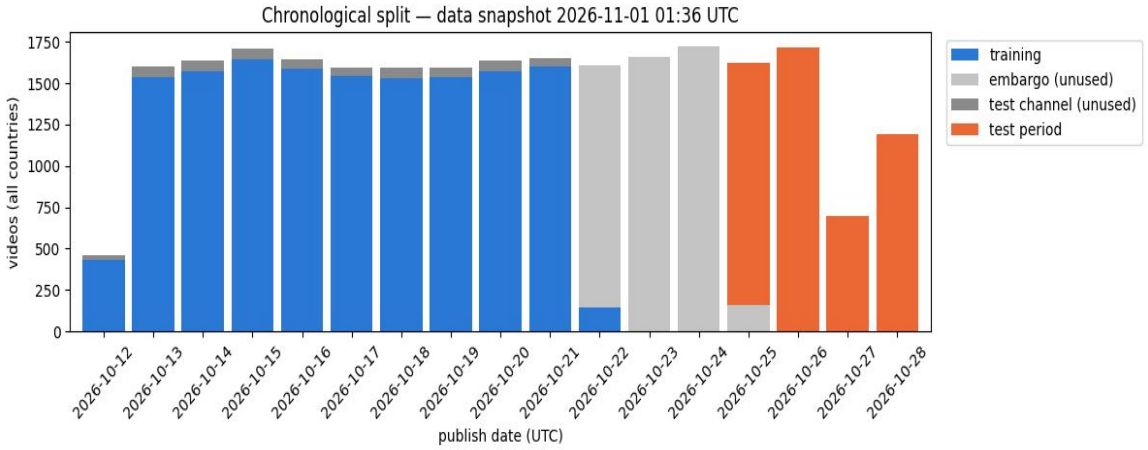
Bundle-level, not causal effects

Raw and log1p engagement were effectively equivalent for XGBoost; raw counts were retained for simplicity.

MODELLING

Evaluation design limits temporal and creator leakage

Development and testing were separated before model comparison.



Source: Modelling.ipynb; frozen chronological split and 72-hour embargo.

Safeguards

Grouped CV by channel

Creators do not appear in both sides of a fold.

Chronological holdout

A 72-hour embargo separates development from test.

Single final evaluation

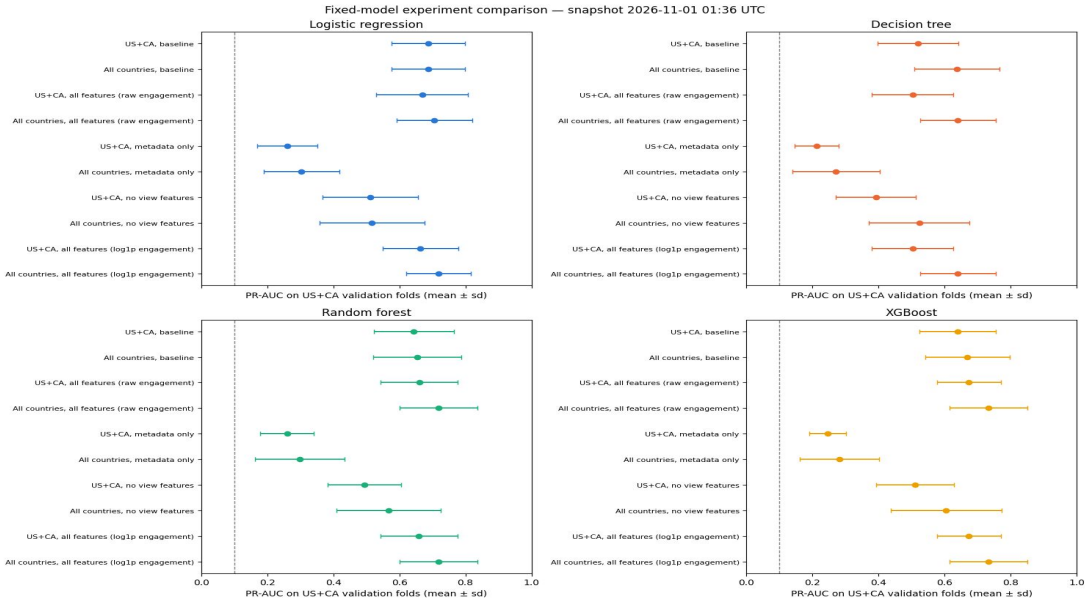
The holdout was evaluated once after model selection.

Target: top 10% of US + Canada Shorts by approximately 72-hour views. Prediction point: six hours.

MODELLING

Development CV favored XGBoost with worldwide training

Candidate models and feature sets were compared using the same grouped folds.



; 3 repeats x 5 grouped folds. Points show mean PR-AUC +/- SD.

Selected development candidate

XGBoost

worldwide training + all raw features

PR-AUC

0.733 +/- 0.119

ROC-AUC

0.936 +/- 0.029

Worldwide training won 87% of paired folds

MODELLING

Final holdout showed a modest, consistent improvement

The selected XGBoost model was compared with ranking videos by six-hour views alone.

PR-AUC

0.685 to **0.759**

+0.074

ROC-AUC

0.880 to **0.913**

+0.033

Precision@10%

0.561 to **0.596**

+0.035

Recall@10%

0.667 to **0.708**

+0.041

Top 10%

57 videos

Final XGBoost

34 of 48 viral videos identified

Six-hour views baseline

32 of 48 identified

n = 562
48 viral

Interpretation: performance improved on all prespecified metrics in this holdout; broader generalization requires additional future-period data.

MODELLING RESULTS

Results

01 / PRIMARY DRIVER

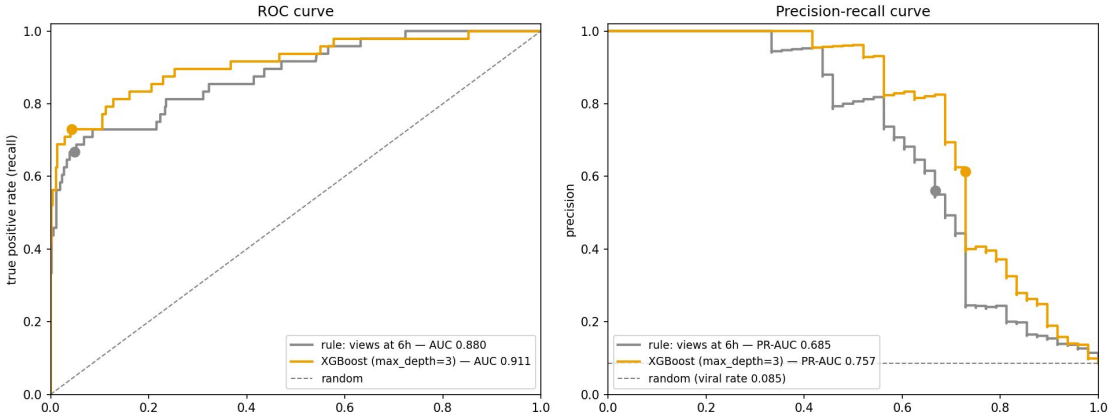
Overall, early views seem to be the biggest indicator of a video becoming viral.

02 / SECONDARY FACTORS

Metadata, Channel Info, and video engagement also play a factor into a video going viral.

CONCLUSION: Timing is key! Hour 6 is the decision point. When a video is flagged, act on it while it's still growing.

Performance Curves (ROC & Precision-Recall)



HOLDOUT METRICS COMPARISON

	PR-AUC	ROC-AUC	precision@10%	recall@10%
rule: views at 6h	0.685	0.880	0.561	0.667
XGBoost (4. All countries, all features (raw engagement))	0.757	0.911	0.614	0.729

Limitations

Sampling Bias

- Could not specifically collect Shorts

YouTube Algorithm

- Affects views and engagement

Missing Data

- Some videos have comments/likes disabled

Future Directions

- Investigate virality prediction before video is published
  - Analyze thumbnails, titles, or tags
- Compare outcomes for large and small channels

Thank You!